



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS ADVANCED

2024 Year 11 Course Assessment Task 2

Friday, 28 June 2024

General instructions

- Working time – 1 hour.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt **all** questions.

SECTION I

- Mark your answers on the answer grid provided (on page 2)

SECTION II

- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #:

Class (please ✓)

☐ 11MAA.1 – Ms Quek

Marker's use only.

QUESTION	1-3	4-5	6-7	8-9	10-11	12-15	16	Total	%
MARKS	3	9	13	7	8	14	6	60	

Section I

3 marks

Attempt Question 1 to 3

Allow approximately 4 minutes for this section

Mark your answers on the answer grid.

Questions

Marks

1. Which of the following is equivalent to 1

$$\frac{\cos(90^\circ - \theta)}{\sin(90^\circ - \theta)}$$

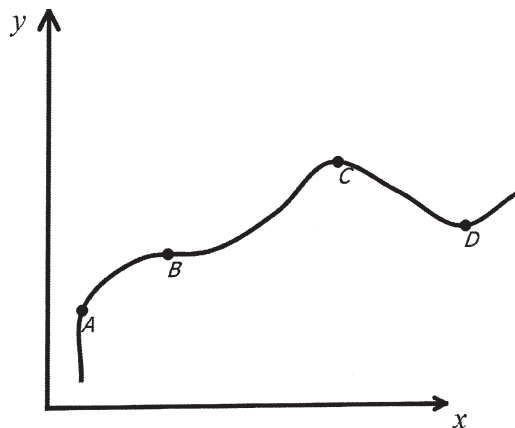
- (A) 1 (B) $\cot \theta$ (C) $\tan \theta$ (D) $-\tan \theta$

2. Evaluate $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x - 1}$ 1

- (A) undefined (B) 0 (C) 1 (D) 4

3. Which of the following points on the graph of $y = f(x)$ has the following properties simultaneously: 1

- $f'(x) = 0$
- $f''(x) = 0$



- (A) A (B) B (C) C (D) D

Answer grid for Section I**1** – ☐ A ☐ B ☐ C ☐ D**2** – ☐ A ☐ B ☐ C ☐ D**3** – ☐ A ☐ B ☐ C ☐ D**Examination continues overleaf...**

Section II

57 marks

Attempt Question 4 to 16

Allow approximately 56 minutes for this section

Write your answers in the space provided.

Question 4 (6 marks)

Differentiate with respect to x

(a) $x^2\sqrt{x}$ **2**

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(b) $(2x^2 + 1)^6$ **2**

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(c) $\frac{1 + 7x}{x^3}$ **2**

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Question 5 (3 marks)

If $f(x) = \frac{1}{x-1}$, show that $f'(x) = -\frac{1}{(x-1)^2}$ by differentiation from first principles. **3**

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Examination continues overleaf...

Question 6 (5 marks)

- (a) **By using the product rule**, show that the derivative of $y = (3x^3 + x)(x - 1)$ is **2**

$$\frac{dy}{dx} = 12x^3 - 9x^2 + 2x - 1$$

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- (b) Hence, find the equation of the normal to the curve $y = (3x^3 + x)(x - 1)$ at the point where $x = 1$. Express your answer in general form $ax + by + c = 0$. **3**

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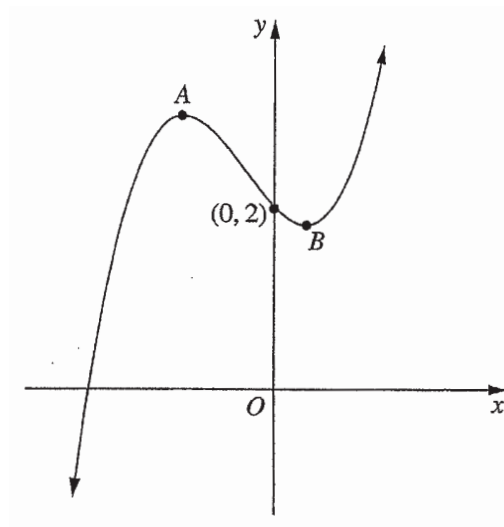
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Question 7 (8 marks)

The graph of $y = x^3 + x^2 - x + 2$ is sketched below.



The points A and B are the turning points.

- (a) Find the coordinates of A and B .

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- (b) For what values of x is the curve concave up?

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Show all mathematical reasons for your answer.

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- (c) For what value(s) of k does the equation $x^3 + x^2 - x + 2 = k$ only has one real solution?

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Examination continues overleaf...

Question 8 (4 marks)

Given $\log_7 2 = 0.36$ and $\log_7 5 = 0.83$, find the values of the following.

(a) $\log_7 0.4$ **2**

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(b) $\log_7 50$ **2**

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Question 9 (3 marks)

In the space below, sketch the graph of $y = \log_5(5x - 5)$, showing all key features including: **3**

- any x intercept(s)
- any asymptote(s)

Examination continues overleaf...

Question 10 (4 marks)

- (a) Solve the following equation for x , correct to 2 decimal places. **2**
Full working out must be shown to gain full marks.

$$0.4^{x+6} = 128$$

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- (b) By considering the graph of $y = 0.4^{x+6}$ and also part (a), find the values of x such that **2**

$$0.4^{x+6} < 128$$

(*Hint: draw a rough sketch of the graph of $y = 0.4^{x+6}$*)

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Examination continues overleaf...

Question 11 (4 marks)

A drug is used to control a medical condition. It is known that the quantity Q of drug remaining in the body after t hours satisfies an equation of the form

$$Q = Q_0 e^{-kt} \quad \text{where } Q_0 \text{ and } k \text{ are constants}$$

The initial dose is a 6 milligrams and after 15 hours the amount remaining in the body is half the initial dose.

- (a) Find the value of Q_0 . **1**

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- (b) Show that the value of k is 0.046, correct to 3 decimal places. **1**

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- (c) Show (with full working out) that it takes approximately 45 hours 12 minutes for one-eighth of the initial dose to remain. **2**

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Examination continues overleaf...

Question 12 (4 marks)

Sydney is 694 nautical miles (M) south and 761 M west of Noumea. **4**

A plane must fly to make a trip from Noumea to Sydney. Calculate the distance and the true bearing of the flight to the nearest nautical miles and degree respectively.

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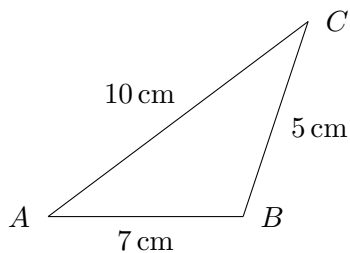
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Examination continues overleaf...

Question 13 (3 marks)

The diagram shows $\triangle ABC$ with sides $AB = 7\text{cm}$, $BC = 5\text{cm}$ and $AC = 10\text{cm}$.



- (a) Show that $\cos A = \frac{31}{35}$ **1**

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- (b) Hence by finding the exact value of $\sin A$, determine the exact area of $\triangle ABC$. **2**

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Examination continues overleaf...

Question 14 (2 marks)

Given $\sin \theta = \frac{3}{4}$ and $\tan \theta < 0$, find the exact value of $\cos \theta$

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Question 15 (5 marks)

(a) Prove that $\frac{2 \cos^2 \theta}{1 - \sin \theta} = 2 + 2 \sin \theta$

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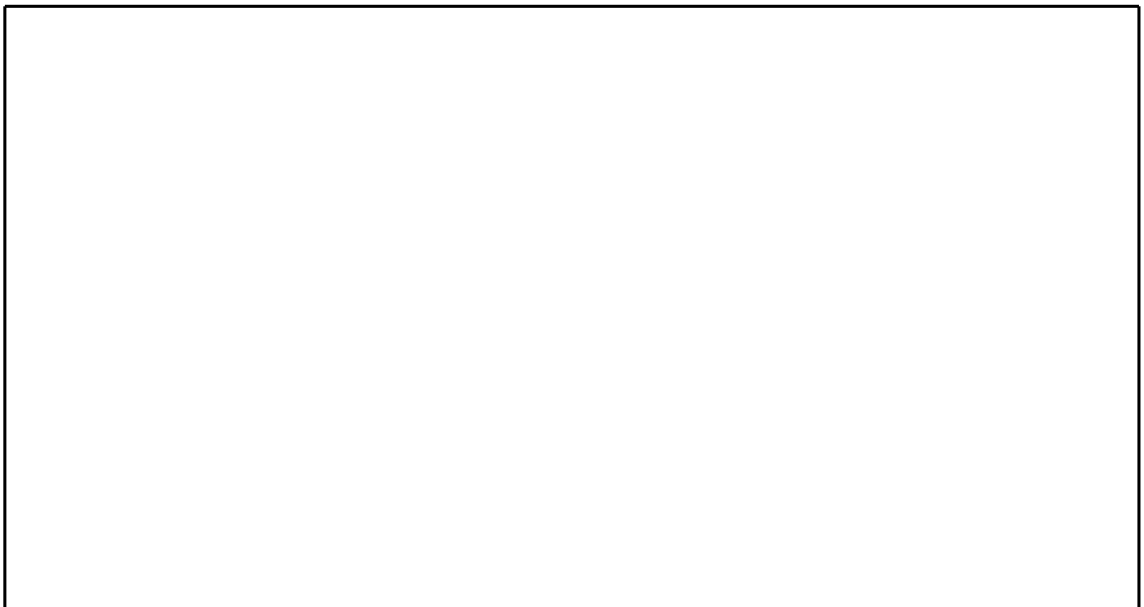
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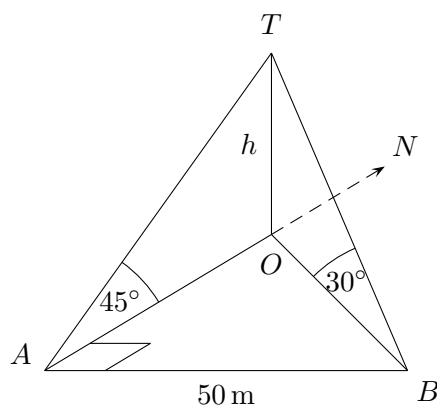
(b) Hence draw a sketch of a function $y = \frac{2 \cos^2 \theta}{1 - \sin \theta}$ in the domain $-90^\circ < \theta < 90^\circ$

3

Examination continues overleaf...

Question 16 (6 marks)

A person is standing at point A , which is due south of a tower OT that is h metres tall. The person walks 50 metres east to point B . The angle of elevation at points A and B to the top of the tower are 45° and 30° respectively.



- (a) Express OA and OB in terms of h . **2**

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- (b) Show that $h = 25\sqrt{2}$ metres. **2**

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- (c) Find the bearing of B from the base of the tower O , correct to the nearest minute. **2**

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End of paper.

Additional writing space



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SECTION I

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SECTION II

- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: ...Sample...solution...

Class (please ✓)

☐ 11MAA.1 – Ms Quek

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Section I

3 marks

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Mark your answers on the answer grid.

Questions

Marks

1. Which of the following is equivalent to 1

$$\frac{\cos(90^\circ - \theta)}{\sin(90^\circ - \theta)} = \frac{\sin \theta}{\cos \theta} = \tan \theta$$

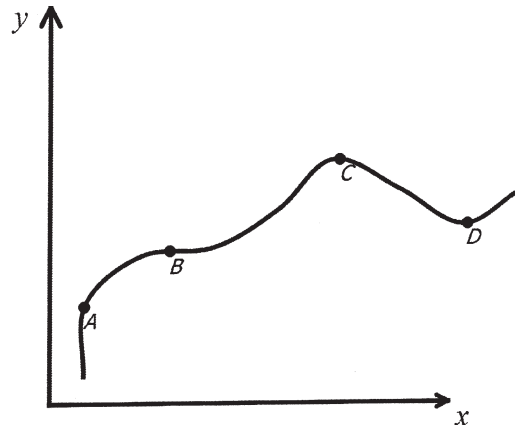
- (A) 1 (B) $\cot \theta$ (C) ☒ $\tan \theta$ (D) $-\tan \theta$

2. Evaluate $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x - 1}$ 1

- (A) undefined (B) 0 (C) 1 (D) ☒ 4

3. Which of the following points on the graph of $y = f(x)$ has the following properties simultaneously: 1

- $f'(x) = 0 \rightarrow$ stationary point
- $f''(x) = 0 \rightarrow$ Horizontal point of inflection



- (A) A (B) ☒ B (C) C (D) D

Answer grid for Section I

- | | | | | |
|-----|-------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 – | ☐ A | ☐ B | ☒ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☒ D |
| 3 – | ☐ A | ☒ B | ☐ C | ☐ D |

Examination continues overleaf...

Section II

57 marks

Attempt Question 4 to 16

Allow approximately 56 minutes for this section

Write your answers in the space provided.

Question 4 (6 marks)

-1 Setting out overall.

Differentiate with respect to x

(a) $x^2\sqrt{x} = x^{2+\frac{1}{2}} = x^{\frac{5}{2}}$ ① $\frac{d}{dx}(x^{\frac{5}{2}}) = \frac{5}{2}x^{\frac{3}{2}}$ ① 2

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(b) $(2x^2 + 1)^6$ 2

$\frac{d}{dx}(2x^2 + 1)^6 = 6(2x^2 + 1)^5 \times 4x$ ①

$= 24x(2x^2 + 1)^5$ ①

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(c) $\frac{1+7x}{x^3} = \frac{1}{x^3} + \frac{7x}{x^3}$ 2

$= x^{-3} + 7x^{-2}$ ①

$\frac{d}{dx}(x^{-3} + 7x^{-2}) = -3x^{-4} + 7x^{-2} \times -2x^{-3}$

$= -\frac{3}{x^4} - \frac{14}{x^3}$ ①

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Question 5 (3 marks)

If $f(x) = \frac{1}{x-1}$, show that $f'(x) = -\frac{1}{(x-1)^2}$ by differentiation from first principles.

3

$$f(x+h) = \frac{1}{x+h-1}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

① for keeping $\lim_{h \rightarrow 0}$ till the end

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h-1} - \frac{1}{x-1}}{h}$$

①

$$= \lim_{h \rightarrow 0} \frac{\frac{x-1 - (x+h-1)}{(x+h-1)(x-1)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x-1-x-h+1)}{h(x+h-1)(x-1)}$$

$$= \lim_{h \rightarrow 0} \frac{-h}{h(x+h-1)(x-1)}$$

$$= -\frac{1}{(x-1)(x-1)}$$

$$= -\frac{1}{(x-1)^2}$$

①

Examination continues overleaf...

Question 6 (5 marks)

- (a) By using the product rule, show that the derivative of $y = (3x^3 + x)(x - 1)$ is **2**

$$\frac{dy}{dx} = 12x^3 - 9x^2 + 2x - 1$$

$$u = 3x^3 + x \quad v = x - 1$$

$$u' = 9x^2 + 1 \quad v' = 1$$

$$\frac{dy}{dx} = (9x^2 + 1)(x - 1) + (3x^3 + x) \quad (1)$$

$$= 9x^3 - 9x^2 + x - 1 + 3x^3 + x$$

$$= 12x^3 - 9x^2 + 2x - 1 \quad (1)$$

- (b) Hence, find the equation of the normal to the curve $y = (3x^3 + x)(x - 1)$ at the point where $x = 1$. Express your answer in general form $ax + by + c = 0$. **3**

$$\text{at } x=1, \quad y = (3(1)^3 + 1)(1 - 1) = 0$$

$$m_{\text{tangent}} = 12(1)^3 - 9(1)^2 + 2(1) - 1 = 4$$

$$\therefore m_{\text{normal}} = -\frac{1}{4}$$

$$\text{Equation of normal: } y - 0 = -\frac{1}{4}(x - 1)$$

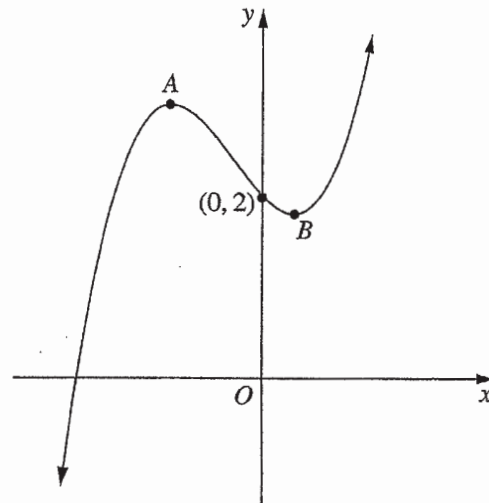
$$-4y = x - 1 \quad (1)$$

$$\therefore x + 4y - 1 = 0 \quad (1)$$

(1) - For leaving answer in general form.

Question 7 (8 marks)

The graph of $y = x^3 + x^2 - x + 2$ is sketched below.



The points A and B are the turning points.

- (a) Find the coordinates of A and B .

4

$$\frac{dy}{dx} = 3x^2 + 2x - 1$$

$$\text{When } \frac{dy}{dx} = 0;$$

$$3x^2 + 2x - 1 = 0 \quad (1)$$

$$(3x - 1)(x + 1) = 0$$

$$x = \frac{1}{3}, -1. \quad (1)$$

$$\text{When } x = \frac{1}{3}, y = \left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^2 - \left(\frac{1}{3}\right) + 2 = \frac{49}{27}$$

$$\text{When } x = -1, y = 3 \quad (1) \text{ - For finding both } y \text{ values}$$

$$\therefore A(-1, 3) \text{ \& } B\left(\frac{1}{3}, \frac{49}{27}\right) \quad (1)$$

- (b) For what values of x is the curve concave up?

2

Show all mathematical reasons for your answer.

$$\text{Concave up when } \frac{d^2y}{dx^2} > 0$$

$$\frac{d^2y}{dx^2} = 6x + 2 \quad \therefore 6x + 2 > 0 \quad (1)$$

$$x > -\frac{1}{3}. \quad (1)$$

- (c) For what value(s) of k does the equation $x^3 + x^2 - x + 2 = k$ only has one real solution?

2

$$k > 3 \quad (1) \text{ or } k < \frac{49}{27} \quad (1)$$

Examination continues overleaf...

Question 8 (4 marks)

Given $\log_7 2 = 0.36$ and $\log_7 5 = 0.83$, find the values of the following.

(a) $\log_7 0.4$

2

$$\begin{aligned} \log_7 0.4 &= \log_7 \frac{2}{5} \\ &= \log_7 2 - \log_7 5 \quad (1) \\ &= 0.36 - 0.83 \\ &= -0.47 \quad (1) \end{aligned}$$

(b) $\log_7 50$

2

$$\begin{aligned} \log_7 50 &= \log_7 (25 \times 2) \\ &= \log_7 5^2 + \log_7 2 \quad (1) \\ &= 2 \log_7 5 + \log_7 2 \\ &= 2 \times 0.83 + 0.36 \\ &= 2.02 \quad (1) \end{aligned}$$

Question 9 (3 marks)

In the space below, sketch the graph of $y = \log_5(5x - 5)$, showing all key features including:

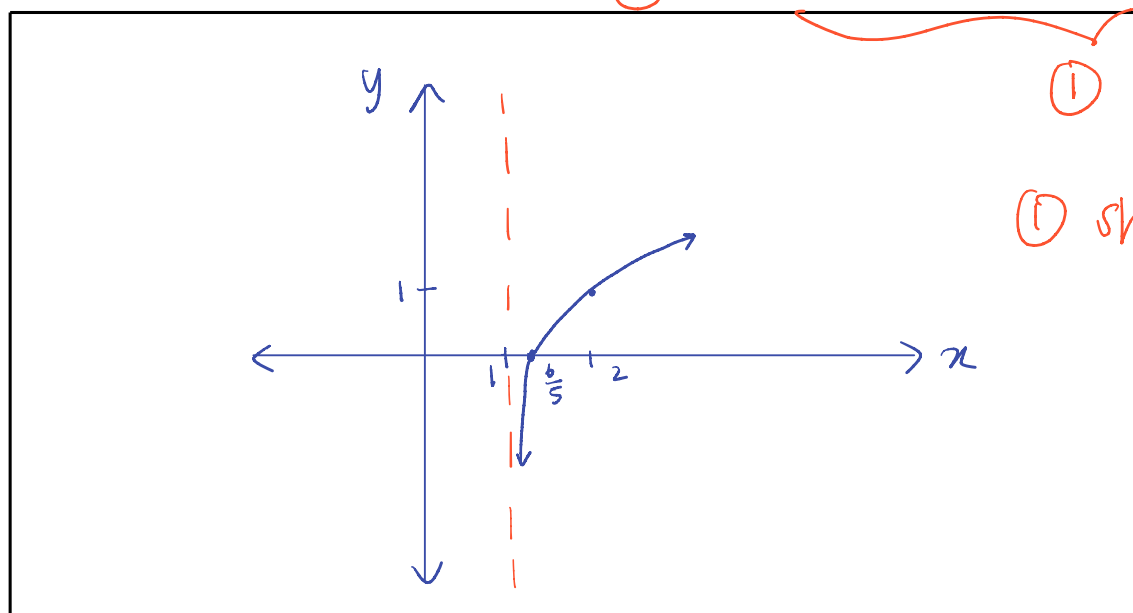
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- any x intercept(s)
- any asymptote(s)

asymptote;
when $5x - 5 = 0$
 $x = 1$

when $x = 2$;
 $y = \log_5(5 \times 2 - 5)$
 $= 1$

x -intercept;
when $5x - 5 = 1$
 $5x = 6$
 $x = \frac{6}{5}$



Examination continues overleaf...

Question 10 (4 marks)

- (a) Solve the following equation for
- x
- , correct to 2 decimal places.

2*Full working out must be shown to gain full marks.*

$$0.4^{x+6} = 128$$

$$\log 0.4^{x+6} = \log 128$$

$$(x+6) \log 0.4 = \log 128$$

$$x+6 = \frac{\log 128}{\log 0.4}$$

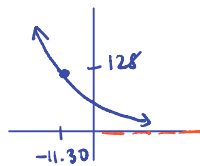
$$x = \frac{\log 128}{\log 0.4} - 6$$

$$= -11.30 \text{ (2dp)}$$

- (b) By considering the graph of
- $y = 0.4^{x+6}$
- and also part (a), find the values of
- x
- such that

2

$$0.4^{x+6} < 128$$

(Hint: draw a rough sketch of the graph of $y = 0.4^{x+6}$)

$$x > -11.30$$

Examination continues overleaf...

Question 11 (4 marks)

A drug is used to control a medical condition. It is known that the quantity Q of drug remaining in the body after t hours satisfies an equation of the form

$$Q = Q_0 e^{-kt} \quad \text{where } Q_0 \text{ and } k \text{ are constants}$$

The initial dose is a 6 milligrams and after 15 hours the amount remaining in the body is half the initial dose.

- (a) Find the value of Q_0 .

1

when $t=0$, $Q=6$

$$6 = Q_0 e^{-k \times 0}$$

$$\therefore Q_0 = 6$$

- (b) Show that the value of k is 0.046, correct to 3 decimal places.

1

when $t=15$, $Q = \frac{1}{2} Q_0$

$$\frac{1}{2} Q_0 = Q_0 e^{-15k}$$

$$\ln \frac{1}{2} = \ln e^{-15k}$$

$$\therefore -15k = \ln \frac{1}{2}$$

$$k = \frac{\ln \frac{1}{2}}{-15}$$

$$= 0.046 \text{ (3dp)}$$

- (c) Show (with full working out) that it takes approximately 45 hours 12 minutes for one-eighth of the initial dose to remain.

2

$t=?$ $Q = \frac{1}{8} Q_0$

$$\frac{1}{8} Q_0 = Q_0 e^{-0.046t}$$

$$\ln \frac{1}{8} = \ln e^{-0.046t}$$

$$-0.046t = \ln \frac{1}{8}$$

$$t = \frac{\ln \frac{1}{8}}{-0.046}$$

$$= 45^{\circ} 12' 18.9''$$

$$\therefore \text{takes approx. 45hrs 12mins.}$$

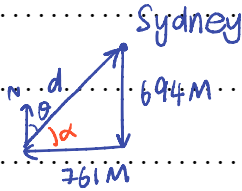
Examination continues overleaf...

Question 12 (4 marks)

Sydney is 694 nautical miles (M) south and 761 M west of Noumea.

4

A plane must fly to make a trip from Noumea to Sydney. Calculate the distance and the true bearing of the flight to the nearest nautical miles and degree respectively.



$$d = \sqrt{694^2 + 761^2} \quad (d > 0) \rightarrow \textcircled{1}$$

$$= 1029.93 \dots$$

$$= 1030 \text{ M}$$

①

$$\tan \alpha = \frac{694}{761}$$

$$\alpha = 42^\circ 21' 48.6''$$

$$= 42^\circ \text{ (nearest degree)}$$

$$\therefore \theta = 90^\circ - 42^\circ$$

$$= 48^\circ$$

①

$\therefore$ bearing from Noumea to Sydney is

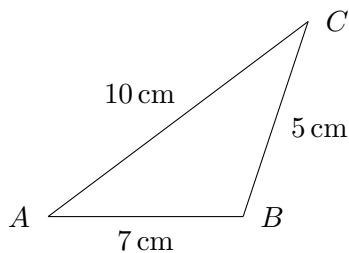
$$048^\circ \text{T}$$

①

Examination continues overleaf...

Question 13 (3 marks)

The diagram shows $\triangle ABC$ with sides $AB = 7\text{cm}$, $BC = 5\text{cm}$ and $AC = 10\text{cm}$.



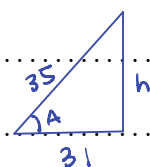
- (a) Show that $\cos A = \frac{31}{35}$

1

$$\begin{aligned}\cos A &= \frac{7^2 + 10^2 - 5^2}{2 \times 7 \times 10} \quad (1) \\ &= \frac{124}{140} \\ &= \frac{31}{35}\end{aligned}$$

- (b) Hence by finding the exact value of $\sin A$, determine the exact area of $\triangle ABC$.

2



$$\begin{aligned}h &= \sqrt{35^2 - 31^2} \\ &= \sqrt{264} \\ &= 2\sqrt{66}\end{aligned}$$

$$\therefore \sin A = \frac{2\sqrt{66}}{35}$$

(1)

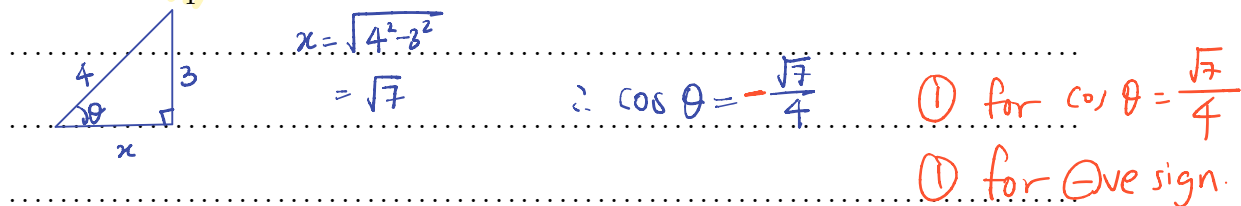
$$\begin{aligned}A_{\triangle ABC} &= \frac{1}{2} \times 7 \times 10 \times \frac{2\sqrt{66}}{35} \\ &= 2\sqrt{66} \text{ units}^2 \quad (1)\end{aligned}$$

Examination continues overleaf...

Question 14 (2 marks)

Given $\sin \theta = \frac{3}{4}$ and $\tan \theta < 0$, find the exact value of $\cos \theta$

2

**Question 15** (5 marks)

(a) Prove that $\frac{2 \cos^2 \theta}{1 - \sin \theta} = 2 + 2 \sin \theta$

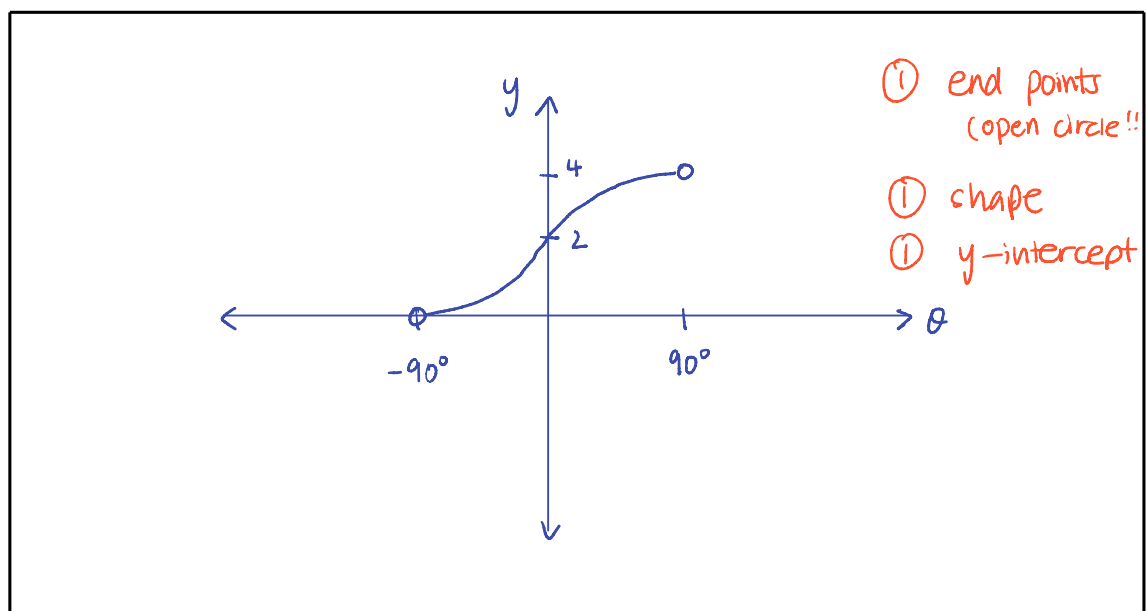
2

$$\begin{aligned}
 \text{LHS} &= \frac{2 \cos^2 \theta}{1 - \sin \theta} \\
 &= \frac{2(1 - \sin^2 \theta)}{1 - \sin \theta} \quad \text{①} \\
 &= \frac{2(1 + \sin \theta)(1 - \sin \theta)}{1 - \sin \theta} \\
 &= 2 + 2 \sin \theta \quad \text{①} \\
 &= \text{RHS}
 \end{aligned}$$

(b) Hence draw a sketch of a function $y = \frac{2 \cos^2 \theta}{1 - \sin \theta}$ in the domain $-90^\circ < \theta < 90^\circ$

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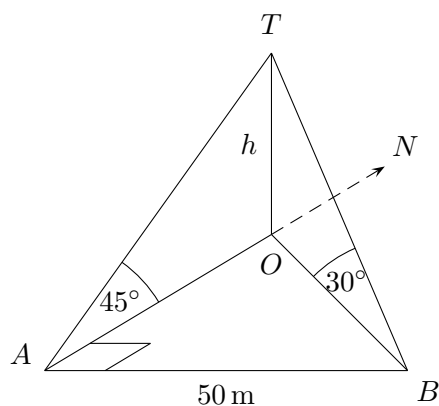
$$\begin{aligned}
 y &= \frac{2 \cos^2 \theta}{1 - \sin \theta} \\
 &= 2 + 2 \sin \theta \\
 a &= 2 \quad n = 1 \quad T =
 \end{aligned}$$



Examination continues overleaf...

Question 16 (6 marks)

A person is standing at point A , which is due south of a tower OT that is h metres tall. The person walks 50 metres east to point B . The angle of elevation at points A and B to the top of the tower are 45° and 30° respectively.



- (a) Express OA and OB in terms of h . 2

$$\begin{aligned} \tan 45^\circ &= \frac{h}{OA} & \tan 30^\circ &= \frac{h}{OB} \\ OA &= \frac{h}{\tan 45^\circ} & OB &= \frac{h}{\tan 30^\circ} \\ &= h \quad (1) & &= \sqrt{3}h \quad (1) \end{aligned}$$

- (b) Show that $h = 25\sqrt{2}$ metres. 2

$$\begin{aligned} \text{In } \triangle AOB, \quad OB^2 &= OA^2 + AB^2 \\ (\sqrt{3}h)^2 &= h^2 + 50^2 \quad (1) \\ 2h^2 &= 50^2 \\ h &= \sqrt{\frac{50^2}{2}} \quad h > 0 \quad \therefore h = \sqrt{1250} \\ &= 25\sqrt{2} \text{ metres} \quad (1) \end{aligned}$$

- (c) Find the bearing of B from the base of the tower O , correct to the nearest minute. 2

$$\begin{aligned} \tan \angle AOB &= \frac{AB}{OA} \\ &= \frac{50}{25\sqrt{2}} \\ \angle AOB &= 54^\circ 44' \text{ (nearest mins)} \quad (1) \\ \therefore \text{bearing of } B \text{ from } O &= 125^\circ 16' \text{ (nearest min)} \quad (1) \end{aligned}$$

End of paper.